

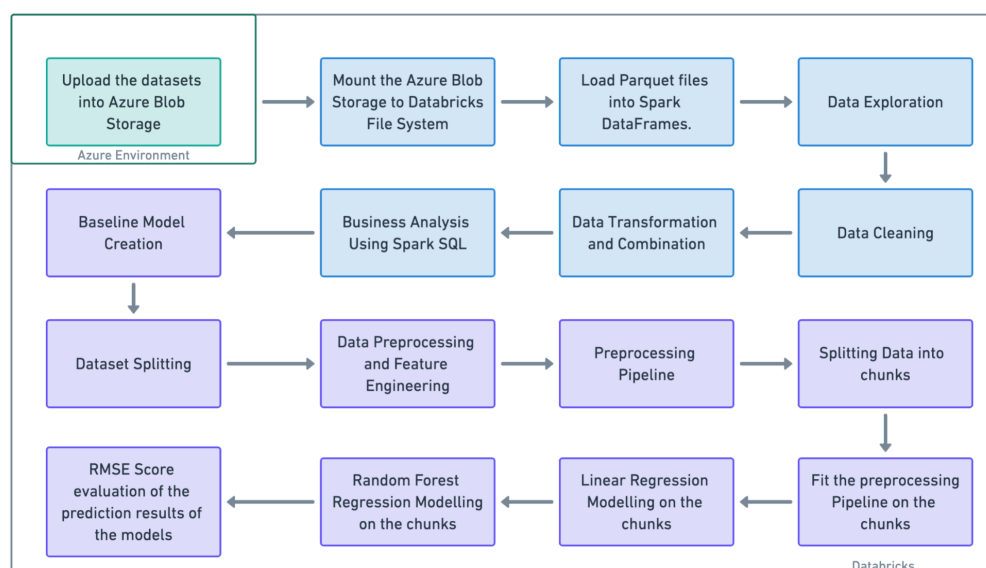
1. Introduction

Using a large amount of data on New York City taxi trips provided by the Taxi and Limousine Commission (TLC), this project aims to process, analyse, and create predictive models. Millions of trip records from yellow and green taxis from 2015 to 2022 are included in the dataset. Every trip record has comprehensive details about the locations of pick-up and drop-off points, travel times, distances travelled, number of passengers, methods of payment, and total amount paid by passengers. The project shows how to use Spark to analyse large-scale datasets and create predictive models, with the ultimate goal of giving drivers and taxi operators useful insights. This is accomplished by creating effective pipelines for data

1.2 Overview of Key Stages

- Load the data into Databricks from Azure Blob Storage to ingest and prepare it for analysis.
- Perform data transformation and cleansing to eliminate unreasonable trips, and get the data ready for additional analysis.
- Using Spark SQL, respond to business enquiries to get insights of taxi operations, such as trip trends, locations, and periods.
- Create machine learning models to forecast a trip's total amount based on the given features using Spark MLlib.
- Utilizing the Root Mean Square Error (RMSE) to compare and assess machine learning models and select the best model based on its performance.

Figure 1: Flowchart of Data processing on a big dataset with Databricks Spark



2. Data Source Exploration

The datasets have been provided by the New York City Taxi and Limousine Commission (TLC), which has made millions of trip records from both yellow and green taxis available to the public. These datasets provide a thorough history of all taxi journeys made in New York City between 2015 and 2022, enabling us to examine travel patterns, payment trends, and passenger behaviour. Every trip record provides a rich dataset for analysis, containing important data such as pick-up/drop-off locations, trip duration, miles travelled, passenger count, and payment details. Parquet format is used to store data for both yellow and green taxis, and it is an effective way to process massive amounts of data.

2. 1. Data Structure

With over 663 million entries for yellow taxis and 66 million for green taxis, the dataset has approximately 729 million rows total for both yellow and green taxis. The taxi datasets in yellow and green share the majority of fields; however, certain columns are different. As an illustration of the distinctions in services, green taxis have 'trip_type' and 'ehail_fee', whereas yellow taxis have 'airport_fee'.

2.1.1 Yellow Taxis: Famous yellow taxis are available to pick up people wherever in New York City. In Manhattan in particular, yellow taxis are more common and usually carry more passengers.

2.1.2 Green Taxis: These taxis are only permitted to pick up clients via street hailing in specific locations, including the outer boroughs.

3. Explanation of the Different Steps

3.1 Data Upload in Azure Cloud Storage

The datasets containing the parquet files for the trip records of yellow and green taxis have been uploaded to Azure cloud storage. To provide delegated access to the resources to be able to use in Databricks, the storage account name, storage account access key, and blob storage container name have been retrieved from the Azure environment.

3.2 Data Ingestion and Preparation

The massive taxi datasets in Parquet format have been loaded into Databricks from Azure Blob Storage using the storage account name, storage account access key, and the blob

container name. To make effective use of Spark for data processing, the datasets were put into the Databricks File System (DBFS).

- Step 1: Mount the Azure Blob Storage Container to the Databricks File System.
- Step 2: Make respective copies of the green and yellow taxi files into DBFS.
- Step 3: Load Parquet files into Spark DataFrames.
- Step 4: Count and verify the number of records for each taxi color using Spark SQL:
 - Green Taxi: 66,200,401 rows
 - Yellow Taxi: 663,055,251 rows
- Step 5: Load a referential CSV file containing location information (such as boroughs and zones) for later use in analysis and feature engineering.

Figure 2: Snippet of Trip Records of Green Taxis

VendorID	lpep_pickup_datetime	lpep_dropoff_datetime	store_and_fwd_flag	RatecodeID	PULocationID	DOLocationID	passenger_count	trip_distance	fare_amount	extra	mta_tax	tip_amount	tolls_amount	ehail_fee	improvement_surcharge	total_amount	payment_type	trip_type	congestion_surcharge
2	2015-05-01T00:24:18.000+0000	2015-05-01T00:28:29.000+0000	N	1	146	146	1	0	2.5	0.5	0.5	0	0	null	0.3	3.8	2	1	null
2	2015-05-01T00:28:15.000+0000	2015-05-01T00:29:00.000+0000	N	1	146	146	1	0	2.5	0.5	0.5	0	0	null	0.3	3.8	2	1	null
2	2015-05-01T00:07:45.000+0000	2015-05-01T00:12:50.000+0000	N	1	255	112	1	1.09	5.5	0.5	0.5	1	0	null	0.3	7.8	1	1	null
2	2015-05-01T00:26:23.000+0000	2015-05-01T00:26:12.000+0000	N	1	255	13	1	5.73	22	0.5	0.5	4.68	0	null	0.3	27.96	1	1	null
2	2015-05-01T00:20:16.000+0000	2015-05-01T00:30:37.000+0000	N	1	80	225	1	2.35	9.5	0.5	0.5	0	0	null	0.3	10.8	2	1	null
2	2015-05-01T00:37:21.000+0000	2015-05-01T00:50:00.000+0000	N	1	37	112	1	2.68	11	0.5	0.5	3.08	0	null	0.3	15.38	1	1	null
2	2015-05-01T00:16:15.000+0000	2015-05-01T00:31:20.000+0000	N	1	255	154	1	4.74	19.5	0.5	0.5	0	0	null	0.3	25.8	2	1	null
2	2015-05-01T00:22:31.000+0000	2015-05-01T01:05:05.000+0000	N	1	17	68	1	6.47	29.5	0.5	0.5	0	0	null	0.3	30.8	2	1	null
2	2015-05-01T00:27:04.000+0000	2015-05-01T00:35:33.000+0000	N	1	112	226	5	2.25	9	0.5	0.5	2.08	0	null	0.3	12.36	1	1	null
2	2015-05-01T00:19:16.000+0000	2015-05-01T00:35:52.000+0000	N	1	255	52	1	4.9	17.5	0.5	0.5	3.76	0	null	0.3	22.56	1	1	null

Figure 3: Snippet of Trip Records of Yellow Taxis

VendorID	lpep_pickup_datetime	lpep_dropoff_datetime	passenger_count	trip_distance	RatecodeID	store_and_fwd_flag	PULocationID	DOLocationID	payment_type	fare_amount	extra	mta_tax	tip_amount	tolls_amount	improvement_surcharge	total_amount	congestion_surcharge	airport_fee
1	2015-01-01T00:11:33.000+0000	2015-01-01T00:16:48.000+0000	1	1	1	N	41	166	1	5.7	0.5	0.5	1.4	0	0	8.4	null	null
1	2015-01-01T00:18:24.000+0000	2015-01-01T00:24:20.000+0000	1	0.9	1	N	166	238	3	6	0.5	0.5	0	0	0	7.3	null	null
1	2015-01-01T00:26:19.000+0000	2015-01-01T00:41:06.000+0000	1	3.5	1	N	238	162	1	13.2	0.5	0.5	2.9	0	0	17.4	null	null
1	2015-01-01T00:45:28.000+0000	2015-01-01T00:53:20.000+0000	1	2.1	1	N	162	353	1	8.2	0.5	0.5	2.37	0	0	11.87	null	null
1	2015-01-01T00:59:21.000+0000	2015-01-01T01:05:24.000+0000	1	1	1	N	238	141	3	6	0.5	0.5	0	0	0	7.3	null	null
1	2015-01-01T00:07:31.000+0000	2015-01-01T00:11:32.000+0000	1	0.8	1	N	239	238	2	5	0.5	0.5	0	0	0	6.3	null	null
1	2015-01-01T00:47:08.000+0000	2015-01-01T00:54:30.000+0000	1	1.1	1	N	238	239	2	7	0.5	0.5	0	0	0	8.3	null	null
1	2015-01-01T00:58:04.000+0000	2015-01-01T01:11:56.000+0000	1	2.9	1	N	238	42	1	12.2	0.5	0.5	2.7	0	0	16.2	null	null
1	2015-01-01T00:29:25.000+0000	2015-01-01T00:37:25.000+0000	2	1.3	1	N	90	125	2	7	0.5	0.5	0	0	0	8.3	null	null
1	2015-01-01T00:39:02.000+0000	2015-01-01T01:02:37.000+0000	2	4.3	1	N	125	141	2	16	0.5	0.5	0	0	0	19.3	null	null

3.3 Comparison of Parquet and CSV Formats

The file sizes of each version have been examined after converting the dataset of Green Taxi 2015 from Parquet to CSV format. The comparison's outcomes were as follows:

- The Green Taxi 2015 Parquet file has a size of 385.81 MB.
- The Green Taxi 2015 CSV file has a size of 2078.17 MB.

Despite having the same data, the CSV file is more than five times larger than the Parquet file. The advantages of utilising Parquet format over CSV for large datasets are demonstrated by this notable difference in file size, particularly when big data processing is involved. Since Parquet is a columnar storage system, information is kept in columns as opposed to rows.

This makes it possible to use extremely effective compression methods that greatly minimise file size. Contrarily, CSV does not support any kind of compression and keeps data as plain text, which leads to significantly bigger file sizes. Large datasets are best served by Parquet since it is a binary format that is optimised for performance in distributed computing settings such as Spark. As CSV files are text-based, they are less effective for large-scale processing and take longer to read and write.

3.4 Data Analysis

The following procedures were used to investigate and examine the 2015–2022 New York City taxi dataset (which includes both green and yellow taxis).

- Top Pick-up Locations:
 - Green Taxis: PULocationID 74 is the most popular pick-up location, with 3,940,683 trips, which is East Harlem North in Manhattan. PULocationIDs 41, 75, and 7 were the next most popular locations; these correspond to neighbourhoods of Manhattan and Queens, where green taxis are mostly found.
 - Yellow Taxis: PULocationID 237 is the most popular pickup location, with 26,803,919 trips, which is Upper East Side South of Manhattan. The concentration of other top locations, such as PULocationID 161, 236 and 186, in central districts like Manhattan is indicative of the extensive usage of yellow taxis in the city centre.

Figure 4: Top Pickup Locations for Green (Left) and Yellow taxis (Right)

PULocationID	count	PULocationID	count
74	3940683	237	26803919
41	3512123	161	24663410
75	3415470	236	24342496
7	3008143	186	22481441
82	2658298	162	22365759

- Top Drop-off Locations:
 - Green Taxis: 2,141,866 trips at DOLocationID 74 in East Harlem North in Manhattan. Like pick-up locations, the majority of green taxi drop-offs occurred in boroughs outside of Manhattan. and Queens.
 - Yellow Taxis: DOLocationID 236 in Upper East Side North of Manhattan is the top drop-off location, with 24,795,550 trips. The patterns of yellow taxi drop-offs and pick-ups are consistent, indicating Manhattan's high usage.

Figure 5: Top Dropoff Locations for Green (Left) and Yellow taxis (Right)

DOLocationID	count	DOLocationID	count
74	2141866	236	24795550
42	2113383	161	24578969
41	1870492	237	23443565
7	1859264	170	20825824
129	1720407	230	19925306

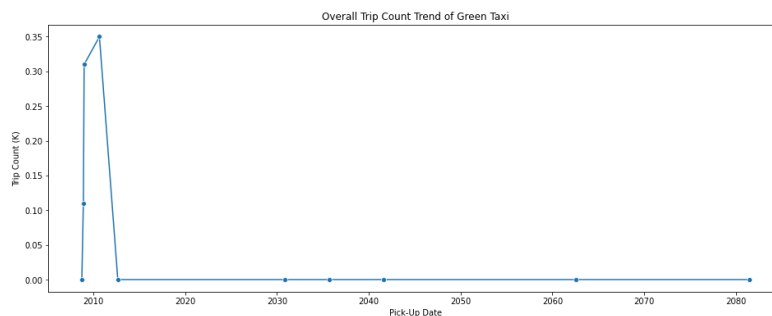
- Outliers in total_amount:
 - Green Taxis: There were inaccuracies or outliers present since the maximum fare was \$989,970.39 and the mean fare was \$15.53. Unreasonable negative fares were also discovered, with a minimum charge of -\$890.3
 - Taxis in yellow: At \$17.23, the average fare was somewhat more expensive. With a maximum fare of \$3,950,611.60 and a minimum fare of -\$2,567.8, there were some severe outliers.

Figure 6: total_amount for Green (Left) and Yellow taxis (Right)

summary		total_amount	summary		total_amount
0	count	66200401	0	count	663055251
1	mean	15.528623431472889	1	mean	17.227453900471343
2	stddev	122.37304907093659	2	stddev	232.4413282762861
3	min	-890.3	3	min	-2567.8
4	max	989970.39	4	max	3950611.6

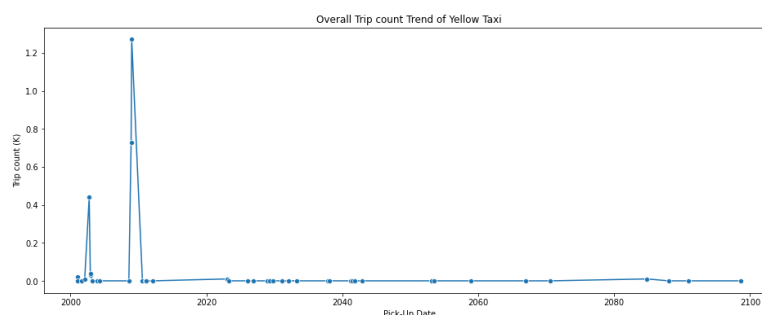
- Unusual dates for the trips:
 - Green Taxis: Despite the fact that green taxis were introduced in 2013, there were journeys recorded between 2010 and 2080. Travels beyond 2022 are clearly anomalies, and a significant surge around 2010 points to potential data input problems.

Figure 7: Odd dates for the trips of green taxis



- Yellow Taxis: Similar problems were found. The dataset included the years 2000 to 2100. An enormous jump in the year 2100 points to inaccurate data.

Figure 8: Odd dates for the trips of yellow taxis



- **Anomalies in Trip Distances:**
 - **Green Taxis:** While the average trip distance was 6.69 miles, there were some anomalies, such as negative mileage and an unreasonable maximum value of 360,068.14 miles.
 - **Yellow Taxis:** A lowest distance of -40.84 million miles and a maximum distance of 59.02 million miles were among the extreme outliers. The average trip distance was slightly lower, at 5.64 miles. These numbers point to errors with data entry.

Figure 9: Trip distances for Green (Left) and Yellow taxis (Right)

summary trip_distance			summary trip_distance		
0	count	66200401	0	count	663055251
1	mean	6.687233460866467	1	mean	5.635196239731811
2	stddev	667.8704347617046	2	stddev	5388.28033007515
3	min	-20329.08	3	min	-4.08401244E7
4	max	360068.14	4	max	5.90166093E7

- **Atypical Passengers count:**
 - **Green Taxis:** A maximum of 48 passengers and a minimum of 0 passenger count have been identified as the irrelevant values.
 - **Yellow Taxis:** With some severe outliers, such as trips with 192 passengers, yellow taxis experienced trips with zero passengers, which is absurd.

Figure 10: Passenger Counts for Green (Left) and Yellow taxis (Right)

summary passenger_count			summary passenger_count		
0	count	64395196	0	count	658895459
1	mean	1.3559085059699174	1	mean	1.6037897508108339
2	stddev	1.0274261472057662	2	stddev	1.2499185992588113
3	min	0.0	3	min	0.0
4	max	48.0	4	max	192.0

3.5 Data Cleaning

A number of data cleaning procedures were used to eliminate inaccurate entries in order to guarantee the accuracy and integrity of the dataset for both green and yellow taxis. The main objectives were to remove unnecessary trips, deal with data errors, and guarantee that all trip records fell within acceptable parameters.

- Trips where the departure time was earlier than the arrival time were deemed erroneous and were eliminated from the database. This was carried out independently for yellow and green taxis:
 - For green taxis: `lpep_dropoff_datetime > lpep_pickup_datetime`
 - For yellow taxis: `tpep_dropoff_datetime > tpep_pickup_datetime`
- Trips were limited to those that took place between January 1, 2015, and December 31, 2022, in order to preserve a valid date range. Records that had pickup or drop-off times that fell outside of this window were deleted.
- The distance travelled and the pick-up and drop-off timings were used to determine the trip's duration and speed. Trips that were unrealistically fast or negative in velocity were eliminated.
 - Green taxis: According to the state speed restriction, trips travelling from 1 to 55 mph were accepted.
 - Yellow taxis: The citywide speed limit in New York City was 1 to 25 mph for all trips.
- Trips that were deemed invalid and eliminated if they lasted longer than two hours or less than one minute. For both yellow and green taxis, only journeys lasting one to two hours were kept.
- Any trip covering less than 0.1 miles or more than 100 miles was eliminated in order to guarantee realistic travel distances. Based on reasonable assumptions for travels inside New York City, these figures were selected.
- Trips with no passengers were considered invalid since it was likely that these records were the product of unusual circumstances or mistakes in data entry. Both green and yellow taxis were only kept for trips involving a minimum of one passenger.

3.6 Data Transformation and Combination

For a unified analysis, it was required to standardise and merge the yellow and green taxi datasets because they had slightly distinct schemas (with some columns being specific to one dataset). This was completed by:

- To guarantee comparable schemas across both datasets, missing columns (`ehail_fee` and `trip_type` for yellow and `airport_fee` for green taxis) were added. Additionally, to keep the record of the taxi color, the column `taxi_color` has been added to both datasets.

- The green and yellow taxi datasets were concatenated into a single DataFrame.
- At each trip's pickup and drop-off locations, borough and zone information were appended by joining the combined dataset with the location referential data.
- To provide efficient access for future research, the aggregated and cleaned dataset was saved as a Parquet file in the DBFS.

4. Business Questions

- **4.1 Analyzing Monthly Trends and Trip Patterns**
 - Total number of trips: Prior to the COVID-19 pandemic that struck in March 2020, the total number of travels remained largely steady. After that, there was a notable fall.
 - The days and hours with most trips during a week: Wednesdays and Tuesdays in particular typically had the most frequent trip counts during the week, with afternoons (6 PM to 7 PM) being the busiest times.
 - Average Passengers and Fare: Over time, there was a minor fluctuation in the average fare each trip, but the average number of passengers per trip remained reasonably stable at about 1.6, along with the average amount paid per passenger (which remained mostly between ~6 and ~10).
 - Impact of the Pandemic: Trips were significantly reduced during the pandemic, remaining low for the majority of 2020–2021 before beginning to rise by 2022.

Figure 11: Snippet of the table for Monthly Trends and Trip Patterns

	year_month	total_trips	most_trip_day_of_week	most_trip_hour	1.2 average_passengers	1.2 average_amount_paid_per_trip	1.2 average_amount_paid_per_passenger
1	2015-01	47032	Sunday	23	1.6505570675284913	13.375910869193167	8.103876644036287
2	2015-01	14838	Tuesday	1	1.5954306510311362	13.326816956463892	8.35311578591692
3	2015-01	87838	Sunday	16	1.6866959630228375	13.62591839522418	8.078467426224394
4	2015-01	89033	Wednesday	15	1.6399874203946851	13.98719991463605	8.52884585618946
5	2015-01	87617	Monday	8	1.5930812513553305	13.692469954459794	8.594960202319111
6	2015-01	97831	Thursday	16	1.6477701342110374	15.25158456930914	9.258893314641058
7	2015-01	45355	Friday	1	1.6444713923492449	13.845329952593854	8.419319434201169
8	2015-01	77945	Monday	15	1.6619411123227916	14.269313361982716	8.585931990116697
9	2015-01	21328	Wednesday	1	1.6306264066016505	13.409503469618153	8.223528955087009
10	2015-01	5830	Tuesday	4	1.5799313893653517	13.86844768439106	8.777879709043521
11	2015-01	83075	Wednesday	12	1.6134697562443574	13.162743785731552	8.158035646339115
12	2015-01	94512	Sunday	14	1.6899441340782122	12.98548925004049	7.683975457048754
13	2015-01	101493	Friday	10	1.6062782654961425	13.354736385760239	8.314086464735466
14	2015-01	115275	Saturday	1	1.7096942094990242	14.26452431143151	8.343319075527145
15	2015-01	62104	Monday	21	1.6403130233157284	12.687330606720987	7.734700893293415

Figure 12: Total number of trips by Year and Month

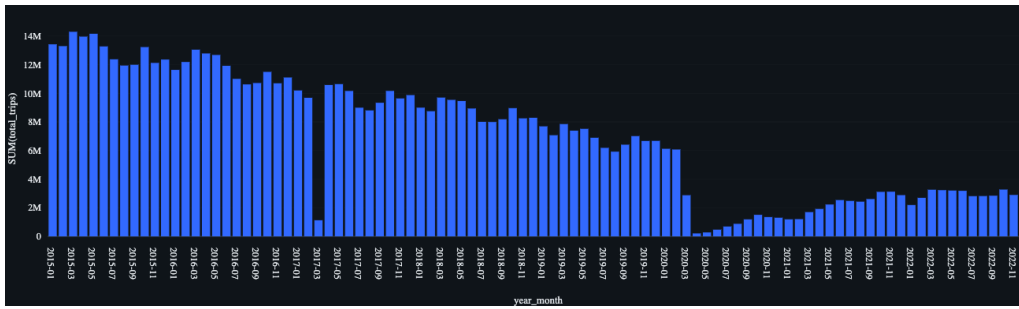


Figure 13: Hour of the day with the most trips by Year and Month

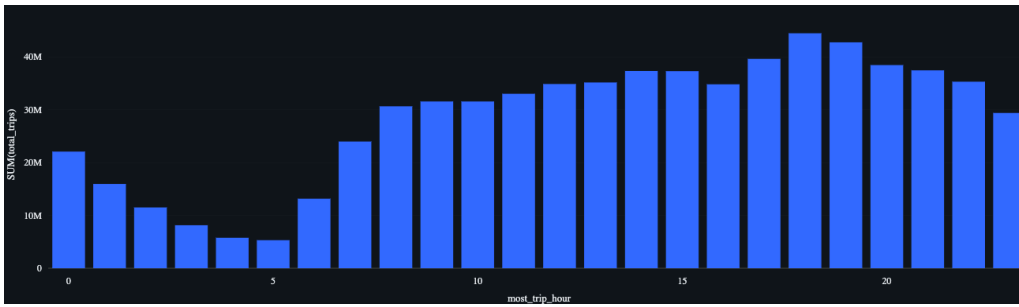


Figure 14: Day of the week with most trips by Year and Month

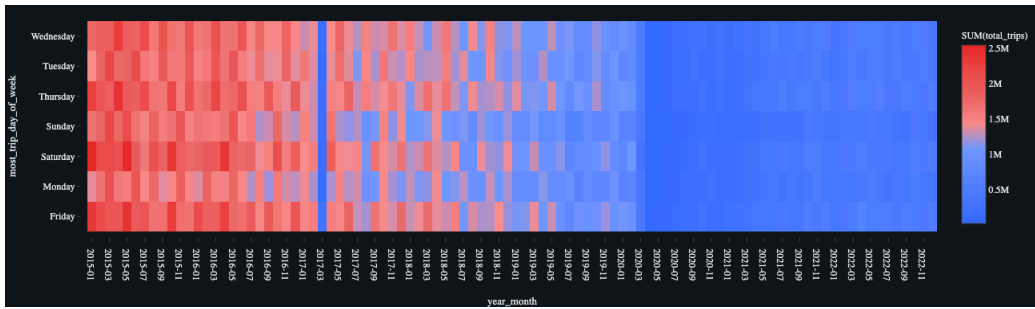


Figure 15: Average amount paid per trip by Year and Month

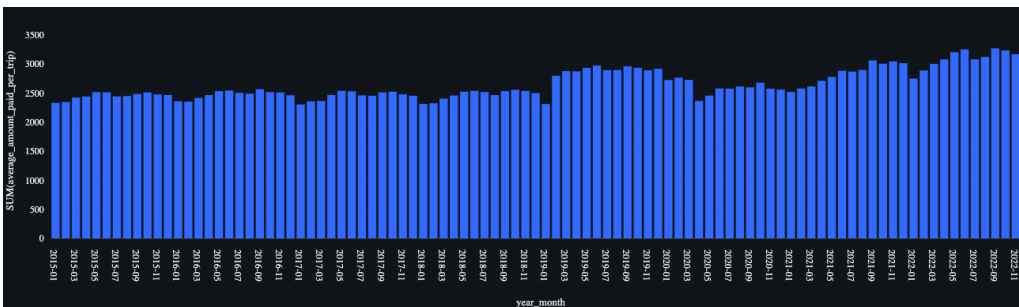


Figure 16: Average amount paid per passenger by Year and Month



• 4.2 Trip Durations, Distances, and Speeds (Yellow and Green Taxis)

- Trip Durations: While both types of taxis could travel a maximum of 120 minutes, yellow taxis' average and median trip times were somewhat greater than those of green taxis. (Average trip durations for green taxi is 13.58 minutes and yellow taxi is 13.93 minutes; Median trip durations is 10.5 minutes for green taxis and 10.92 minutes for yellow taxis.)
- Trip Distances: It's possible that because green taxis operate in boroughs where longer travels are more usual, their average trip distances are higher (4.73 km) than those of yellow taxis (4.17 km).
- Speeds: Compared to yellow taxis (17.29 km/h), green taxis recorded higher average speeds (20.14 km/h), which may indicate that there is less traffic congestion or more usage of highways in the areas where they operate.

Figure 17: Table of Average, Median, Minimum and Maximum trip duration, distance and speed by taxi colors

taxi_color	avg_trip_duration...	median_trip_duration...	min_trip_duration...	max_trip_duration...	avg_trip_distance...	median_trip_distance...	min_trip_distance...	max_trip_distance...	avg_speed...	median_speed...	min_speed...	max_speed...
green	13.58	10.5	1	120	4.73	3.06	0.16	157.78	20.14	16.31	1.61	88.51
yellow	13.93	10.92	1	120	4.17	2.81	0.16	80.19	17.29	16.04	1.61	40.23

Figure 18: Average, Median, Minimum and Maximum Trip Duration for each taxi color



Figure 19: Average, Median, Minimum and Maximum Trip distance for each taxi color

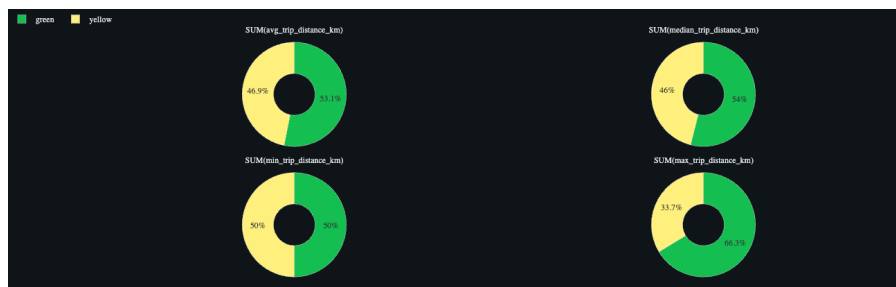
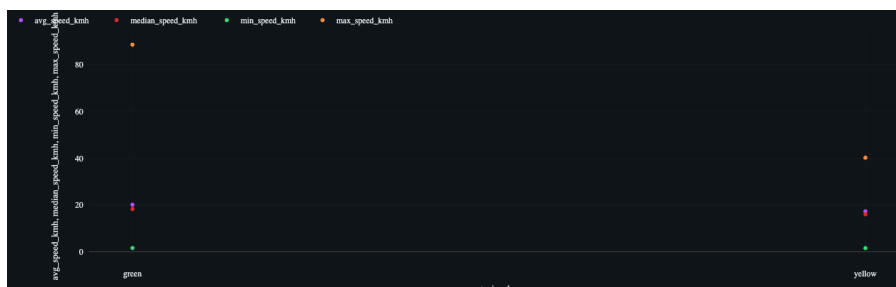


Figure 20: Average, Median, Minimum and Maximum Trip speed for each taxi color



4.3 Trips by Location, Time, and Day for each taxi color

To determine when and where taxis are busiest, this query looks into the relationship between pick-up and drop-off locations, days of the week, and times of day:

- Total Number of Trips: The most trips were seen on weekdays, especially on Mondays and Fridays. Sunday had fewer trips, most likely as a result of fewer commuters on holidays.
- Trip Distances: Within boroughs like Manhattan, shorter trips were more usual than longer ones across boroughs, such as Manhattan to Queens.
- Busiest Hours: The most trips occurred in the afternoon, between 2 and 4 PM, but late-night and early-morning trips were significantly less common.

Figure 21: Snippet of the table for Trips by Location, Time, and Day for each taxi color

taxi_color	pickup_borough	dropoff_borough	year_month	day_of_week	pickup_hour	total_trips	avg_trip_distance_km	avg_amount_paid_per_trip	total_amount_paid
yellow	Bronx	Bronx	2020-09	Friday	21	3	2.13	7.15	21.44
yellow	Bronx	Manhattan	2020-09	Friday	21	6	6.77	20.42	122.51
yellow	Brooklyn	Brooklyn	2020-09	Friday	21	22	3.3	12.04	264.84
yellow	Brooklyn	Manhattan	2020-09	Friday	21	13	7.46	25.18	327.36
yellow	Brooklyn	Queens	2020-09	Friday	21	4	6.1	14.65	68.6
yellow	Manhattan	Bronx	2020-09	Friday	21	56	12.06	30.11	1686.44
yellow	Manhattan	Brooklyn	2020-09	Friday	21	208	9.91	28.92	6014.88
yellow	Manhattan	Manhattan	2020-09	Friday	21	6083	3.33	14.53	88413.84
yellow	Manhattan	Queens	2020-09	Friday	21	149	10.03	28.56	4255.43
yellow	Manhattan	Unknown	2020-09	Friday	21	2	6.89	30.53	61.05
yellow	Queens	Bronx	2020-09	Friday	21	1	11.59	30.42	30.42
yellow	Queens	Brooklyn	2020-09	Friday	21	14	13.97	32.78	458.95
yellow	Queens	Manhattan	2020-09	Friday	21	31	12.75	36.92	1144.45
yellow	Queens	Queens	2020-09	Friday	21	62	6.75	17.49	1084.2
yellow	Unknown	Brooklyn	2020-09	Friday	21	1	5.47	20.76	20.76

Figure 22: Trips by each hour



Figure 23: Trips by each day

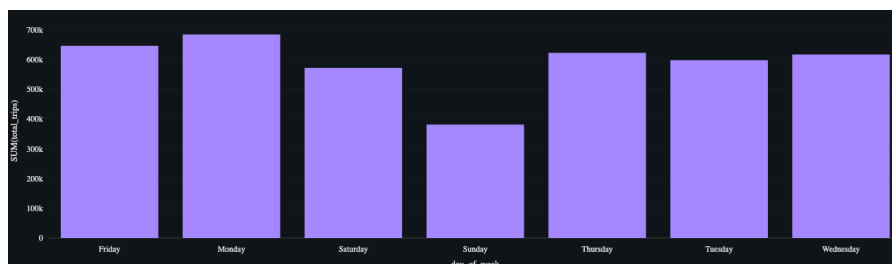
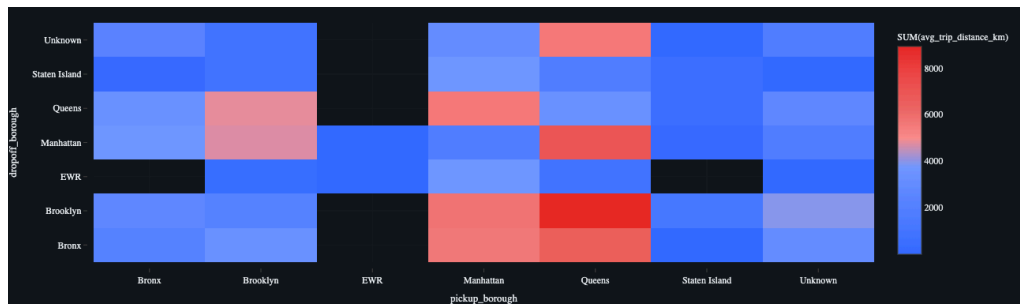


Figure 24: Average distance by each day



Figure 25: Heatmap of pickup and dropoff borough by average distance



4.4 Percentage of Tipped Trips

The percentage of trips that resulted in tips was 63.51%, suggesting that tipping is a prevalent practice.

Figure 26: Table of Percentage of trips where drivers received tips

1^2_3 total_trips	1^2_3 trips_with_tips	1.2 percentage_of_tipped_trips
679709515	431689188	63.51

4.5 Percentage of Trips with Tips of at least \$5

Among the trips with tips, the percentage of trips that received a tip of \$5 or more was only 9.55%, indicating a lower frequency of larger tips.

Figure 27: Table of Percentage of tipped trips with \$5 or more

1^2_3 total_tipped_trips	1^2_3 trips_with_tips_of_5_or_more	1.2 percentage_tips_of_5_or_more
431689188	41212691	9.55

4.6 Classifying Trips by Duration and Analyzing Speed and Distance per Dollar

This query determines the average speed and distance per dollar by classifying trips into several bins according to their duration:

- Speed: Potentially as a result of more highway driving, longer trips, lasting more than 60 minutes—tended to have greater speeds.
- Distance per Dollar: Longer trips typically yield higher mileage per dollar; journeys lasting at least 60 minutes offer the highest value, at 0.49 km per dollar.

Figure 28: Table of Classification of each trip into bins of durations

	1^2_3 duration_bin	1.2 avg_speed_kmh	1.2 avg_distance_per_dollar
1	At least 60 mins	22.25	0.49
2	From 10 mins to 20 mins	16.55	0.25
3	From 20 mins to 30 mins	18	0.29
4	From 30 mins to 60 mins	21.43	0.35
5	From 5 mins to 10 mins	16.77	0.21
6	Under 5 Mins	19.22	0.16

Figure 29: Average Speed by Trip Duration Bins

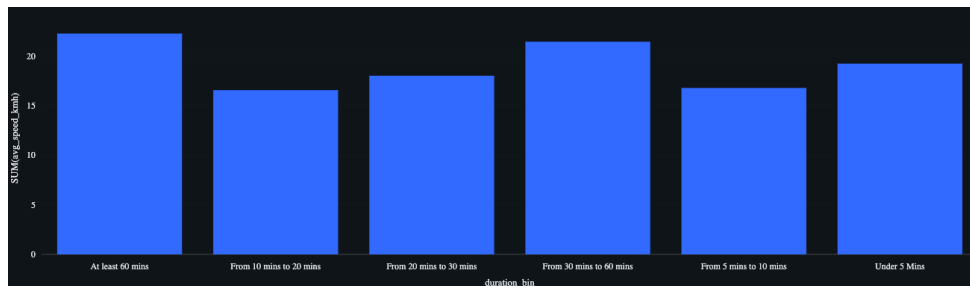
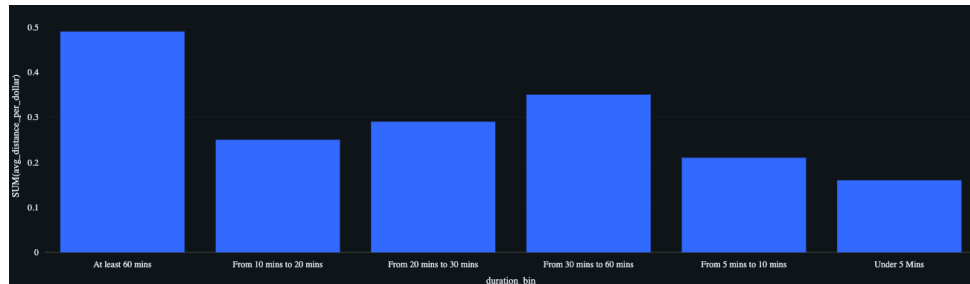


Figure 30: Average Distance per Dollar by Trip Duration Bins



● 4.7 Which Duration Bin is Most Profitable for Taxi Drivers?

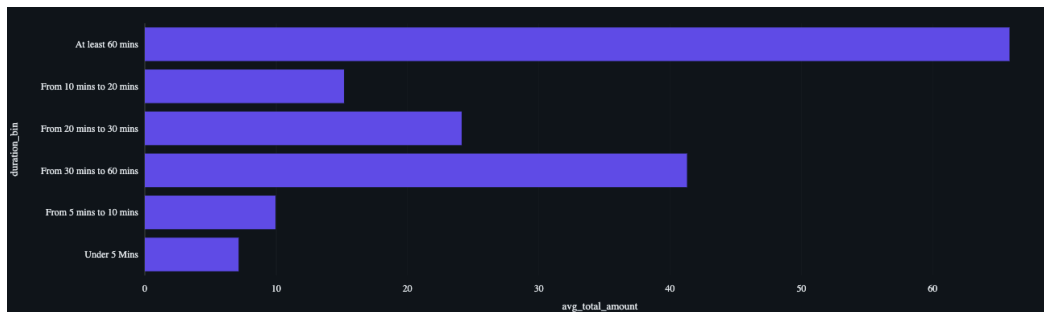
The duration bin with the highest average total amount paid per trip has been determined as the one that a taxi driver should aim for in order to maximise their income. This study will help drivers understand which trip durations are most profitable. At \$65.85 per trip, journeys lasting at least 60 minutes produced the greatest average total amount, followed by trips lasting between 30 and 60 minutes, at \$41.29 each trip.

- Due to the higher charges resulting from the combined distance and time-based fare calculation, longer trips are more profitable because they cover bigger distances.
- Fewer detours or traffic jams may occur during these trips, allowing for longer stretches of uninterrupted driving. This raises average speed and, hence, fare potential.
- Many of these lengthy trips are possibly the result of highway driving or travel between boroughs (such as from/to airports or Manhattan to Queens), which raises speeds and increases fares.

Figure 31: Table of average total amount paid per trip in bins of durations

	duration_bin	avg_total_amount
1	At least 60 mins	65.85
2	From 30 mins to 60 mins	41.29
3	From 20 mins to 30 mins	24.13
4	From 10 mins to 20 mins	15.17
5	From 5 mins to 10 mins	9.95
6	Under 5 Mins	7.14

Figure 32: Average Total Amount paid per Duration Bin



To optimise their income, taxi drivers should prioritise trips that last for a minimum of sixty minutes. These trips offer the most value in terms of distance travelled per dollar, in addition to having the highest overall fare.

5. Machine Learning

To forecast the total amount of a taxi trip based on the cleaned combined dataset, the development and assessment of machine learning models is the main goal of this segment. The procedure comprised building a baseline model, training several machine learning models with various dataset segments, and contrasting the results of Linear Regression and Random Forest Regressor models. Based on training duration, complexity, and RMSE, the optimal model has been chosen.

5.1 Baseline Model Creation

The baseline model predicts each taxi ride using the average total amount for all trips. This offered a fundamental benchmark for assessing the effectiveness of more sophisticated machine learning algorithms.

- The average total sum is \$15.62.
- RMSE at baseline: 194.05

The baseline RMSE is high because it does not take into consideration trip-specific variables like passenger count or distance.

5.2 Data Splitting

The dataset has been divided by training (all data except October to December 2022) and testing (October to December 2022).

5.3 Feature Engineering and Preprocessing

- **Features:** “passenger_count” , “trip_distance” , “payment_type” , “PU_Borough” , “DO_Borough”, and “taxi_color” are chosen for prediction in both the Linear Regression and Random Forest models. The cost may vary depending on the number of passengers on shared journeys or additional extras. Since taxi fares are mostly determined by the distance travelled, “trip_distance” is an important component. The mode of payment (cash or credit, for example) may have an effect on the transaction as a whole and tipping behaviour. The location affects the fare and trip distance; some boroughs may have fixed surcharges (like airport fees) or traffic patterns that affect the overall amount. The operating zones of green and yellow taxis can have an impact on the fare structure, trip distance, and surcharges. On the other hand, IDs, such as “VendorID”, have no direct bearing on fare prediction. The time of day is not important because time-based fees (such as rush hour pricing) have not been employed. Usually set at a fixed value, these surcharges, such as “mta_tax”, “extra”, “congestion_surcharge”, and “ehail_fee” would not increase the generalisability of the model. Including “trip_duration_hours” could result in overfitting and redundancy because of their strong correlation with “trip_distance”.
- **Preprocessing Pipeline**
 - **StringIndexer:** To create numerical indices from category columns such as PU_Borough and DO_Borough.
 - **OneHotEncoder:** For the numerical encoding of categorical characteristics.
 - **VectorAssembler:** To create a single vector for model training by combining all of the characteristics.

5.4 Data Processing in Chunks

As the dataset is massive in size, it has been divided into four types of chunks to avoid overwhelming the computational resources. To fit the Linear Regression and Random Forest models, each chunk was processed independently.

- **Chunk 1:** Small fraction ($1/50000$) of the dataset.
- **Chunk 2:** Medium fraction ($1/10000$) of the dataset.
- **Chunk 3:** Large fraction ($1/1000$) of the dataset.
- **Chunk 4:** Largest fraction ($1/100$) of the dataset.

5.5 Model Training and Evaluation

- **Linear Regression Model:** A simple regression model using input features to estimate the total amount.
- **Random Forest Regressor:** A tree-based ensemble model called the Random Forest Regressor constructs several decision trees in order to provide predictions.
- **Evaluation Metric:** The Root Mean Squared Error (RMSE) has been computed for each model to assess performance.

5.6 Results

The results show that the Linear Regression model outperformed the Random Forest model consistently, obtaining lower RMSE values in every chunk. By having lower RMSE values, both the models beat the baseline model's RMSE (194.05) across all the chunks.

Figure 33: RMSE values for Linear Regression and Random Forest models across all chunks

Chunk	Linear Regression RMSE	Random Forest RMSE
0 Chunk 1	6.782220	7.874491
1 Chunk 2	6.534028	7.857379
2 Chunk 3	6.671989	7.634046
3 Chunk 4	6.898009	7.839862

Figure 34: RMSE Comparison between Linear Regression and Random Forest models across all chunks

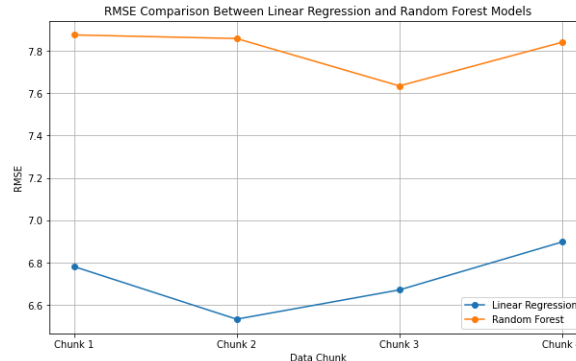


Figure 35: Prediction results (top 5 rows) of Linear Regression (left) and Random Forest (right) models on Chunk 4 (Largest fraction)

features	total_amount	prediction
(23, [0,1,2,9,16,2...]	17.8	13.434083546550916
(23, [0,1,2,9,16,2...]	11.76	9.046465083457418
(23, [0,1,2,9,16,2...]	21.95	12.743622384614167
(23, [0,1,2,9,16,2...]	15.36	13.399560488454078
(23, [0,1,2,9,17,2...]	12.3	10.096252232376532

features	total_amount	prediction
(23, [0,1,2,9,16,2...]	17.8	12.587716274713866
(23, [0,1,2,9,16,2...]	11.76	11.980646056523188
(23, [0,1,2,9,16,2...]	21.95	12.587716274713866
(23, [0,1,2,9,16,2...]	15.36	12.587716274713866
(23, [0,1,2,9,17,2...]	12.3	10.296819506442553

Due to its higher level of complexity, the Random Forest model required a lot longer to train, especially on larger chunks. The Linear Regression model, being simpler, completed training faster and was more efficient in handling larger datasets.

Table 1: Model Comparison (Processing Time) for each model on the different chunks

Chunk	Linear Regression Time	Random Forest Time
Chunk 1	12 minutes	14 minutes
Chunk 2	12 minutes	18 minutes
Chunk 3	12 minutes	20 minutes
Chunk 4	14 minutes	29 minutes

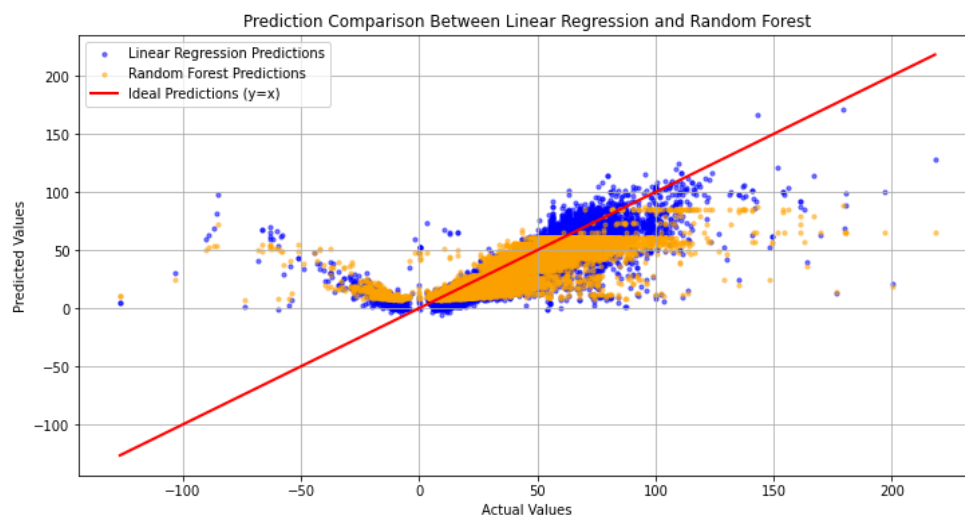
5.7 Best Model Selection

The best model has been determined to be the **Linear Regression** model based on the RMSE, processing time, and model complexity.

- Accuracy (RMSE): With lower RMSE values, better predictions have been consistently obtained using Linear Regression as opposed to Random Forest.
- Processing Time: Especially for larger chunks, the linear regression training time was noticeably faster.
- Model Complexity: On smaller datasets, Linear Regression is less likely to overfit since it is a simpler model with fewer hyperparameters to adjust. Although more complex and time-consuming, the Random Forest model did not perform better than Linear Regression.

The scatter plot (below) contrasts the actual values with the predictions made by the Linear Regression and Random Forest models. The ideal predictions are represented by the red line. Better overall performance is indicated by the plot, which displays that Linear Regression forecasts (blue points) are typically closer to the actual values, particularly in the mid-range.

Figure 36: Scatter Plot of Model Comparison



6. Issues Faced and Their Solutions

- **Cluster Termination or Crashes:** The Databricks cluster's frequent termination was one of the primary issues. This occurred when the enormous dataset overloaded the computational capacity or when the cluster was idle for over sixty minutes. To address this matter, I put the following tactics into practice:
 - **Saving Results Periodically:** I made sure that every output and intermediate result was stored in DBFS as a Parquet file. I was able to save time and cost by not having to restart the entire procedure and reload the data at a later time.
- **Handling Large Datasets:** Due to limited computer resources, processing the entire dataset at once was not possible and sometimes caused crashes or lengthy execution periods. I tackled this issue by doing the following:
 - **Partitioned the Dataset:** I divided the dataset into four types of chunks, namely small, medium, large, and larger, and handled each section independently. Individual Linear Regression and Random Forest models were trained and assessed using each chunk. By not overloading the cluster with the entire dataset, this method made it possible for the model training process to run without going over memory constraints.
- **Limited Cloud Storage Credits:** Since there is a limited amount of credits remaining on my Azure Cloud Storage, I did the following:
 - **Copying Data to DBFS:** I copied the referential CSV file and the datasets (yellow and green taxi data) into DBFS to save cost on cloud storage. This allowed me to work with the data stored locally on Databricks without continually relying on Azure Blob Storage.

7. Conclusion

This project effectively illustrated how to process, analyse, and create machine learning models on big datasets using Databricks Spark. A sequence of procedures including data transformation, data cleansing, and feature engineering allowed for the effective handling of the New York City taxi data from 2015 to 2022. With the use of machine learning models, such as Random Forest and Linear Regression, it is possible to anticipate the entire cost of a trip with some degree of accuracy. In the end, the model that balanced efficiency and accuracy the best was the Linear Regression model.